

Fluid

Mechanics

## The Basics

Let  $\rho: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}$  be the **density**,  $\vec{u}: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$  the **velocity** of a flow. We denote by  $x: \mathbb{R} \rightarrow \mathbb{R}^3$  a **pathline**, i.e. a trajectory of a particle in the flow.

If  $\phi: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}$  is a field, then we denote its total time derivative

$$\frac{D\phi}{Dt} := \frac{\partial}{\partial t} \phi(x(t), t) = \frac{\partial \phi}{\partial t} + \dot{x} \cdot \nabla \phi = \frac{\partial \phi}{\partial t} + \vec{u} \cdot \nabla \phi$$

and call it the **material derivative**.  $\vec{u} \cdot \nabla \phi$  is sometimes called the **advective**.

Let the **mass** of a flow in a region be  $M = \int_V \rho dV$ .

Then

$$\frac{\partial M}{\partial t} + \underbrace{\int_{S=\partial V} \rho \vec{u} \cdot d\vec{S}}_{\text{net flow of mass}} \stackrel{\text{divergence theorem}}{=} \int_V \rho dV + \int_V \nabla \cdot (\rho \vec{u}) dV = \int_V \rho + \nabla \cdot (\rho \vec{u}) dV.$$

We get **conservation of mass** if

$$\rho + \nabla \cdot (\rho \vec{u}) = 0,$$

or equivalently

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} = 0.$$

We will for the most part assume that the fluid is **incompressible**, i.e. that

$$\frac{D\rho}{Dt} = 0,$$

implying that  $\nabla \cdot \vec{u} = 0$ , i.e. that fluid flow is **solenoidal**, or **divergence free**. Usually, we will even have  $\dot{\rho} = \nabla \rho = 0$ , i.e. constant density. In the general case, the fluid is called **stratified**. Note that sound wave flow is not solenoidal.

If  $\nabla \cdot \vec{u} = 0$  then  $\vec{u} = \nabla \times \vec{A}$ . If the flow is independent of the  $z$ -direction, we can write  $\vec{A} = (0, 0, \Psi(x, y, t))$ , where we call  $\Psi$  the **stream function**. Note that lines of constant  $\Psi$  are **streamlines**, i.e. paths along  $\vec{u}$  for constant  $t$ .

## The Euler equation

The Euler equations are the fluid analogue of  $F=ma$ . The momentum inside  $V$  is  $\int_V \rho \vec{u} dV$ .

For the momentum flux at  $\partial S$ , note that  $\partial S$  sweeps out  $\vec{u} \cdot \vec{n} \delta t \delta S = \vec{u} \cdot \vec{\Delta S} \delta t$  in time  $\delta t$ . Hence, the flux is  $\delta t \int_S (\rho \vec{u}) \vec{u} \cdot \vec{\Delta S}$ .

We get

$$\begin{aligned} \text{force} &= \frac{d}{dt} \int_V \rho \vec{u} dV + \int_S (\rho \vec{u}) \vec{u} \cdot \vec{\Delta S} \\ &\stackrel{\text{divergence theorem}}{=} \int_V \left( \rho \frac{D\vec{u}}{Dt} + \rho \frac{\partial \vec{u}}{\partial t} \right) dV + \int_V \nabla \cdot (\rho \vec{u} \vec{u}) dV \\ &\stackrel{\nabla \cdot \vec{u} = 0}{=} \int_V \rho \frac{D\vec{u}}{Dt} dV + \int_V \rho \frac{\partial}{\partial x_j} (u_i u_j) dV \\ &= \int_V \rho \left( \frac{\partial \vec{u}}{\partial t} + u_i \frac{\partial}{\partial x_j} u_j + u_j \frac{\partial}{\partial x_j} u_i \right) dV \\ &= \int_V \rho \left( \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) dV = \int_V \rho \frac{D\vec{u}}{Dt} dV. \end{aligned}$$

*Annotations:*  
 -  $\vec{u} \cdot \vec{n} \delta t \delta S = \vec{u} \cdot \vec{\Delta S} \delta t$ : final vector direction  
 -  $\frac{\partial}{\partial x_j} u_i u_j$ : has components  $\rho u_i u_j$ ,  $\nabla$  applies to the second index  
 -  $\frac{\partial \vec{u}}{\partial t}$ :  $\frac{\partial \vec{u}}{\partial t}$  since  $\nabla \cdot \vec{u} = 0$   
 -  $\frac{\partial}{\partial x_j} u_i u_j$ : above definition, but for vectors

The most important force acting on the fluid acts via pressure

$$F_{\text{pressure}} = -P \vec{n} \delta S,$$

where  $P: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}$ . Note that  $F_{\text{pressure}}$  is isotropic, i.e. the same in all directions. We get

$$\int_V \rho \frac{D\vec{u}}{Dt} dV = - \int_S P \vec{\Delta S} + \text{other forces}.$$

We will assume that the other forces, like gravity, act on volume, so we get a force density  $\vec{f}: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$  s.t.

$$\text{other forces} = \int_V \vec{f} dV.$$

By the divergence theorem, we can write

$$\int_V \rho \frac{D\vec{u}}{Dt} dV = \int_V (-\nabla P + \vec{f}) dV.$$

We get the Euler equation

$$\rho \frac{D\vec{u}}{Dt} = -\nabla P + \vec{f}.$$

With viscosity, these will later become the Navier-Stokes equations. Fluids without viscosity are called ideal.

Let  $\vec{f} = \rho \vec{g}$  be gravity. If the fluid is at rest, i.e.  $\vec{u} = 0$ , we get  $\nabla P = \rho \vec{g}$ ,

i.e.  $P = P_0 - \rho g z$ . This is hydrostatic pressure. We can adjust  $P$  and then just ignore gravity.

If a volume is submerged in the fluid then the fluid exerts

$$F = - \int_S P \vec{\Delta S} = - \int_V \nabla P dV = - \int_V \rho \vec{g} dV$$

on it, which is the weight of the displaced water. This is Archimedes' principle.

Recall Euler's equation

$$\rho \left( \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) = \rho \frac{D\vec{u}}{Dt} = -\nabla P + \vec{f}.$$

Applying the identity  $\vec{u} \times (\nabla \times \vec{u}) = \frac{1}{2} \nabla (\vec{u} \cdot \vec{u}) - (\vec{u} \cdot \nabla) \vec{u}$ , we get

$$\rho \left( \frac{\partial \vec{u}}{\partial t} + \frac{1}{2} \nabla |\vec{u}|^2 - \vec{u} \times (\nabla \times \vec{u}) \right) = -\nabla P + \vec{f}.$$

Multiplying both sides by  $\vec{u}$ , we get

$$\rho \vec{u} \cdot \frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \left( \frac{1}{2} \rho |\vec{u}|^2 + P \right) = \vec{u} \cdot \vec{f}.$$

If  $\vec{f}$  is conservative, i.e.  $\vec{f} = -\nabla \Phi$ , then this becomes

$$\frac{1}{2} \rho \frac{\partial |\vec{u}|^2}{\partial t} + \vec{u} \cdot \nabla \left( \frac{1}{2} \rho |\vec{u}|^2 + P + \Phi \right) = 0,$$

and pulling in  $\vec{u}$  and using  $\nabla \cdot \vec{u} = 0$ ,

$$\frac{1}{2} \rho \frac{\partial |\vec{u}|^2}{\partial t} + \nabla \cdot (\vec{u} H) = 0,$$

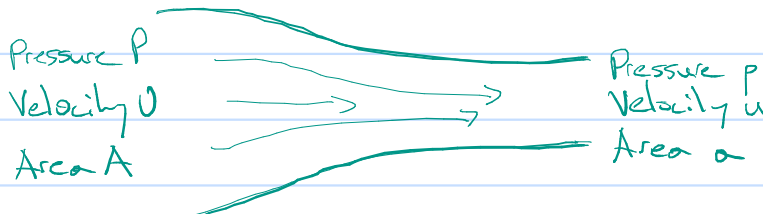
where  $H = \frac{1}{2} \rho |\vec{u}|^2 + P + \Phi$ .

This equation is conservation of energy. If the fluid is steady, i.e.  $\frac{\partial \vec{u}}{\partial t} = 0$ , we get Bernoulli's theorem

$$\vec{u} \cdot \nabla H = 0,$$

i.e.  $H$  is constant along streamlines.

Example:



By conservation of mass,  $UA = ua$ . By Bernoulli's theorem,

$$\frac{\rho}{2} U^2 + P = \frac{\rho}{2} u^2 + p,$$

so we get  $p = P + \frac{\rho}{2} U^2 \left( 1 - \frac{A^2}{a^2} \right)$ .

In particular,  $u = U \frac{A}{a}$  increases while  $p$  decreases.

## Vorticity

Let's look at the derivatives  $\partial_i u_j$ . Note that there are 8 independent derivatives, as  $\nabla \cdot \mathbf{u} = 0$ . They can be decomposed as a symmetric tensor

$$E_{ij} = \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

called **rate of strain**, and an antisymmetric tensor

$$\Omega_{ij} = \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right),$$

which contains the same information as

$$\vec{\omega} = \nabla \times \vec{u}.$$

$\vec{\omega}$  is called the **vorticity**. Its integral curves are **vortex lines** and are perpendicular to streamlines.

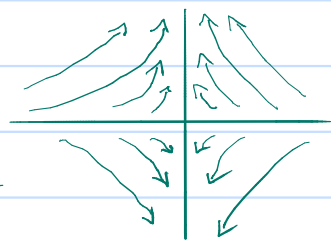
**Example:** A rigid object rotating with angular velocity  $\Omega$  about the origin has velocity  $\vec{u} = \Omega \times \vec{x}$ , hence vorticity

$$\vec{\omega} = \nabla \times (\Omega \times \vec{x}) = 2\Omega.$$

**Example** ( $\vec{u} = (-x, y, 0)$ ): We have

$$\nabla \cdot \vec{u} = 0, \quad \vec{\omega} = 0, \quad E = \begin{pmatrix} -1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The rate of strain shows that the flow is squeezed in the  $x$ -direction and stretched in the  $y$ -direction.



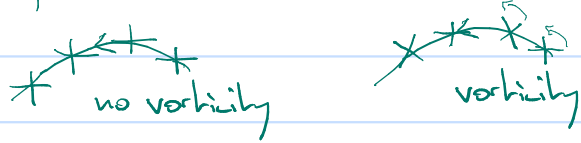
**Example** ( $\vec{u} = (-y, x, 0)$ ): We have

$$\nabla \cdot \vec{u} = 0, \quad E = 0, \quad \vec{\omega} = (0, 0, 2).$$

**Example** ( $\vec{u} = f(r)(-y, x, 0)$ ): We have

$$\nabla \cdot \vec{u} = 0, \quad E = 0, \quad \vec{\omega} = (0, 0, \frac{1}{r} \frac{d}{dr}(r^2 f(r))).$$

In particular, for  $f(r) = \frac{1}{r^2}$ ,  $\vec{\omega} = 0$ . The vorticity tells about local swirling, i.e. about rotation of particles in the fluid themselves.



Since  $\nabla \cdot \vec{u} = 0$ , we can write  $\vec{u} = \nabla \times \vec{A}$ . Hence,  $-\vec{\omega} = \nabla^2 \vec{A}$ . This is the Poisson equation which can be solved using Green's function:

$$\vec{A}(\vec{x}, t) = \frac{1}{4\pi} \int_V d^3x' \frac{\vec{\omega}(\vec{x}', t)}{|\vec{x} - \vec{x}'|},$$

and taking the curl, we get

$$\vec{u}(\vec{x}, t) = \frac{1}{4\pi} \int_V d^3x' \frac{\vec{\omega}(\vec{x}', t) \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}.$$

So the velocity field is sourced by vorticity. This is the **Biot-Savart law**.

To see how vorticity evolves, we recall

$$\rho \frac{\partial \vec{u}}{\partial t} + \frac{1}{2} \rho \nabla |\mathbf{u}|^2 = \rho \vec{u} \times \vec{\omega} - \nabla P - \nabla \Phi.$$

Applying the curl and using that  $\nabla \times (\nabla \cdot) = 0$ , we get

$$\begin{aligned} \frac{\partial \vec{\omega}}{\partial t} &= \nabla \times (\vec{u} \times \vec{\omega}) \\ &= (\nabla \cdot \vec{\omega}) \vec{u} + (\vec{\omega} \cdot \nabla) \vec{u} - (\nabla \cdot \vec{u}) \vec{\omega} - (\vec{u} \cdot \nabla) \vec{\omega}, \end{aligned}$$

hence

$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla) \vec{u},$$

the **vorticity equation**. Taking the product with  $\vec{u}$ , we get

$$\frac{1}{2} \frac{D|\mathbf{u}|^2}{Dt} = \vec{\omega} \cdot (\vec{\omega} \cdot \nabla) \vec{u} = w_i w_j \frac{\partial u_i}{\partial x_j}.$$

Since  $w_i w_j$  is symmetric, this picks out the strain of the flow

$$\frac{1}{2} \frac{D|\mathbf{u}|^2}{Dt} = \vec{\omega} \cdot E \vec{\omega}.$$

We call the integral

$$\Gamma = \oint_C \vec{u} \cdot d\vec{x} = \int_S \vec{\omega} \cdot d\vec{S}$$

the **circulation** of a flow around the closed curve  $C$ . Let  $C(t)$  be a **material curve**, i.e. its points follow the flow. Then

$$\begin{aligned} \frac{D\Gamma}{Dt} &= \oint_{C(t)} \frac{D\vec{u}}{Dt} \cdot d\vec{x} + \vec{u} \cdot \frac{D(d\vec{x})}{Dt} \left\{ \frac{D(d\vec{x})}{Dt} \cdot \nabla \right\} \vec{u} \\ &= \oint_{C(t)} \underbrace{\frac{1}{\rho} \oint_{C(t)} (-\nabla P - \nabla \Phi) \cdot d\vec{x}}_{=0} \cdot d\vec{x} \quad \text{Euler equation} \\ &= \oint_{C(t)} \vec{u} \cdot (d\vec{x} \cdot \nabla) \vec{u} = \oint_{C(t)} u_i \frac{\partial u_i}{\partial x_j} d\vec{x}^j \\ &= \frac{1}{2} \oint_{C(t)} \nabla (\vec{u} \cdot \vec{u}) \cdot d\vec{x} = 0, \end{aligned}$$

i.e. the circulation around a loop does not change if we follow the loop with the flow. This is **Kelvin's circulation theorem**.

An additional conserved quantity is the **helicity**

$$h = \int d^3x \vec{u} \cdot \vec{\omega}.$$

We have

$$\frac{\partial \vec{u}}{\partial t} = -\vec{u} \cdot \nabla \vec{u} - \frac{1}{\rho} \nabla (P + \Phi), \quad \frac{\partial \vec{\omega}}{\partial t} = -\vec{u} \cdot \nabla \vec{\omega} + \vec{\omega} \cdot \nabla \vec{u},$$

so

$$\frac{\partial (\vec{u} \cdot \vec{\omega})}{\partial t} = \nabla \cdot \left( \frac{1}{2} |\mathbf{u}|^2 \vec{\omega} - (\vec{u} \cdot \vec{\omega}) \vec{u} - \frac{1}{\rho} (P + \Phi) \vec{\omega} \right).$$

To get a physical understanding of  $h$ , take two interlinked vortex rings  $w_1, w_2$  with circulations  $\Gamma_1, \Gamma_2$ , resp. Write

$$\begin{aligned} h &= \int_{V_1} d^3x \vec{u} \cdot \vec{\omega}_1 + \int_{V_2} d^3x \vec{u} \cdot \vec{\omega}_2 \\ &= \underbrace{\oint_{C_1} \vec{u} \cdot d\vec{x}_1}_{= \int_{S_1} \vec{\omega} \cdot d\vec{S}_1} \underbrace{\int_{S_1} \vec{\omega}_1 \cdot d\vec{S}_1}_{\Gamma_1} + \dots \\ &= n \Gamma_2 \quad \text{linking number} \\ &= 2n \Gamma_1 \Gamma_2 \end{aligned}$$

## Potential flows in 3D

We now look at steady ( $\frac{\partial \vec{u}}{\partial t} = 0$ ), incompressible ( $\nabla \cdot \vec{u} = 0$ ), irrotational ( $\nabla \times \vec{u} = 0$ ) flows, which are called **potential flows**. By irrotationality,

$$\vec{u} = \nabla \phi.$$

By incompressibility,  $\nabla^2 \phi = 0$ .

Example (Flow around a sphere): For  $q \in \mathbb{R}$ , let

$$\phi = -\frac{q}{r}.$$

Then  $\vec{u} = \frac{q}{r^2} \hat{r}$ .

This satisfies Poisson's equation outside of the origin:

$$\nabla^2 \phi = 4\pi q \delta(\vec{x}).$$

Other solutions have the form in spherical coordinates  $(r, \theta, \varphi)$

$$\phi(r, \theta) = \sum_{n=0}^{\infty} \left( A_n r^n + \frac{B_n}{r^{n+1}} \right) P_n(\cos \theta).$$

The radial solution above corresponds to  $u=0$ . For  $u=1$ ,

$$\phi(r, \theta) = \left( Ar + \frac{B}{r^2} \right) \cos \theta.$$

The first term is  $\phi(x, y, z) = Az$ , a constant flow in the  $z$ -direction. For the second term, consider a dipole

$$\tilde{\phi}(r) = \frac{q}{r} - \frac{q}{|\vec{r} + \vec{d}|}.$$

For  $r \gg |\vec{d}|$ , we have

$$\frac{1}{|\vec{r} + \vec{d}|} \approx \frac{1}{r} + \vec{d} \cdot \nabla \frac{1}{r} = \frac{1}{r} - \frac{\vec{d} \cdot \vec{r}}{r^3},$$

so  $\tilde{\phi}(r) \approx q \frac{\vec{d} \cdot \vec{r}}{r^3}$ .

Let  $\vec{d} = d\hat{z}$ , so  $\vec{d} \cdot \vec{r} = dr \cos \theta$ , and let  $d \rightarrow 0$  with  $qd$  fixed. Then

$$\lim \tilde{\phi}(r) = \frac{qd}{r^2} \cos \theta,$$

The second term in  $\phi(r, \theta)$ .

We have  $\vec{u} = \frac{\partial \phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \hat{\theta}$ ,

with components

$$u_r = \frac{\partial \phi}{\partial r} = \left( U - \frac{2B}{r^3} \right) \cos \theta$$

$$u_\theta = \frac{\partial \phi}{\partial \theta} = - \left( U + \frac{B}{r^3} \right) \sin \theta.$$

Note that  $\vec{u}$  vanishes when  $r^3 = \frac{2B}{U}$ . Hence, this flow, restricted to the outside of the ball with  $r^3 = \frac{2B}{U}$ , corresponds to the flow around the ball!

Let's boost by  $U$ , so that instead of the fluid moving, the sphere moves inside a stationary fluid. By Bernoulli's principle and irrotationality,  $H = \frac{1}{2} \rho |\vec{u}|^2 + P$

is constant along streamlines. Asymptotically,

$$H = \frac{1}{2} \rho U^2 + P_\infty,$$

while on the sphere's surface,

$$H = \frac{\rho}{8} U^2 \sin^2 \theta + P(\theta).$$

Hence, the pressure on the surface of the sphere is

$$P(\theta) = P_\infty + \frac{1}{2} \rho U^2 \left( 1 - \frac{9}{4} \sin^2 \theta \right).$$

Note that the pressure in the front of the sphere ( $\frac{\pi}{2} < \theta \leq \pi$ ) is the same as in the back ( $0 \leq \theta < \frac{\pi}{2}$ )! This is **d'Alembert's Paradox**. It will be resolved with viscosity.

Example (A rising bubble): Consider

$$\phi(r, \theta) = -\frac{UR^3}{2r^2} \cos \theta,$$

a sphere of radius  $R$  moving with speed  $U$ . The kinetic energy of the fluid is

$$\begin{aligned} T_{\text{fluid}} &= \frac{1}{2} \rho \int_{r>R} (\nabla \phi)^2 dV \\ &= \frac{1}{2} \rho \int_{r>R} (\nabla \cdot (\phi \nabla \phi) - \phi \nabla^2 \phi) dV \\ &\stackrel{\text{divergence theorem}}{=} \frac{1}{2} \rho \int \phi \vec{u} \cdot d\vec{S} \\ &= \frac{1}{4} \rho R^3 U^2 \int_0^{2\pi} d\varphi \int_0^\pi d\theta \cos^3 \theta \sin \theta \\ &= \frac{\pi}{3} \rho R^3 U^2 \end{aligned}$$

So the total kinetic energy is

$$T = \frac{1}{2} MU^2 + T_{\text{fluid}} = \frac{1}{2} M_{\text{eff}} U^2,$$

with effective mass  $M_{\text{eff}} = M + \frac{2\pi}{3} \rho R^3$ . ← half the mass of the displaced fluid

If  $M=0$  then  $M_{\text{eff}} = \frac{2\pi}{3} \rho R^3$ . In this case, raising the bubble, i.e. lowering the fluid around it, by some height  $z$ , gives a potential energy of

$$V = - \left( \frac{4\pi}{3} \rho R^3 \right) g z. \quad \leftarrow \text{mass of the displaced water}$$

The energy is then

$$E = \frac{1}{2} M_{\text{eff}} U^2 - 2 M_{\text{eff}} g z.$$

In particular,  $\dot{U} = 2g$ , the acceleration is twice the gravitational acceleration.

## Potential flows in 2D

We work again with flows of the form  $\vec{u} = \nabla\phi$  with  $\nabla^2\phi = 0$ . We have

$$\nabla^2\phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2\phi}{\partial\theta^2}.$$

Example:

$$\phi = q \log \frac{r}{r_0} \Rightarrow u = \frac{q}{r} \hat{r} \Rightarrow \nabla^2\phi = 2\pi q \delta(\vec{x})$$

$$\phi = \frac{\Gamma}{2\pi} \theta \Rightarrow u = \frac{\Gamma}{2\pi r} \hat{\theta} \quad (\text{note: } \phi \text{ is well-defined up to } \Gamma)$$

Example (Circulation around a cylinder): The general solution to the Laplace equation is

$$\phi(r, \theta) = A_0 \log r + B_0 \theta + \sum_{n=1}^{\infty} \left( A_n r^n \cos(n\theta + \alpha_n) + \frac{B_n}{r^n} \cos(n\theta + \beta_n) \right).$$

For  $n=1$ ,

$$\phi(r, \theta) = U \left( r + \frac{R^2}{r} \right) \cos \theta,$$

so

$$\vec{u} = \frac{\partial\phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial\phi}{\partial\theta} \hat{\theta}$$

$$= U \left( 1 - \frac{R^2}{r^2} \right) \cos \theta \hat{r} - U \left( 1 + \frac{R^2}{r^2} \right) \sin \theta \hat{\theta}.$$

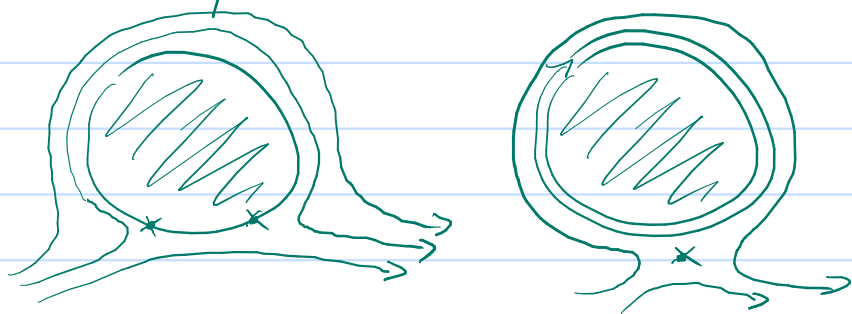
We add the circulation  $\frac{\Gamma}{2\pi} \theta$  from above. We find the stagnation points, where  $\vec{u} = 0$ , at

$$r = R, \quad \Gamma = 4\pi UR \sin \theta.$$

This has solutions only when  $|\Gamma| < 4\pi UR$ . In that case, there are two stagnation points. We find another stagnation point at

$$\theta = \frac{\pi}{2}, \quad r^2 - \frac{\Gamma r}{2\pi U} + R^2 = 0$$

which has a solution only when  $|\Gamma| > 4\pi UR$ .



Let us do the pressure calculation. We have

$$H = \frac{1}{2} \rho U^2 + P_{\infty} = \frac{1}{2} \rho \left( \frac{\Gamma}{2\pi R} - 2U \sin \theta \right)^2 + P(\theta),$$

so

$$P(\theta) = P_{\infty} + \frac{1}{2} \rho U^2 (1 - 4 \sin^2 \theta) + \frac{U \Gamma \rho}{\pi R} \sin \theta - \frac{\Gamma^2 \rho}{8\pi^2 R^2}.$$

We have  $F_z = \int_0^{2\pi} P(\theta) R \cos \theta d\theta = 0$ ,

as with d'Alembert's paradox. But now,

$$F_x = - \int_0^{2\pi} P(\theta) R \sin \theta d\theta = - \frac{U \Gamma \rho}{\pi} \int_0^{2\pi} \sin^2 \theta d\theta = - U \Gamma \rho.$$

There is lift/downforce!

## The Navier-Stokes equation

To get from the Euler equation to the Navier-Stokes equation, we need to add **viscosity**, a force coming from internal friction. For **Newtonian fluids**, which we consider here, this force is linear in  $\vec{u}$ .

The viscosity cannot depend on  $\vec{u}$  directly, because the equation has to stay invariant under boosts, i.e. constant changes to  $\vec{u}$ . The only first derivative that we could use is the vorticity  $\vec{\omega} = \nabla \times \vec{u}$ . But it is even w.r.t.  $\vec{x}$  while each other term is odd. The second derivatives are  $\nabla^2 \vec{u}$  and  $\nabla(\nabla \cdot \vec{u})$ , the second of which vanishes. We get

$$\vec{f}_{\text{viscous}} = \mu \nabla^2 \vec{u},$$

where  $\mu$  is called **dynamic (shear) viscosity**. We get the **Navier-Stokes equation**

$$\rho \frac{D\vec{u}}{Dt} = -\nabla P + \mu \nabla^2 \vec{u} + \vec{f}.$$

We call

$$\nu = \frac{\mu}{\rho}$$

the **kinematic viscosity**.

For boundary conditions, we will assume the "no pass" condition

$$\vec{n} \cdot \vec{u} = 0,$$

where  $\vec{n}$  is the normal field to the boundary, or

$$\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{U},$$

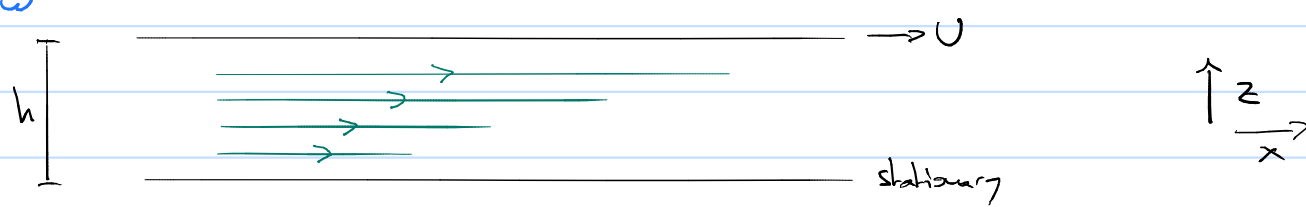
if the boundary moves with velocity  $\vec{U}$ . We will also assume the "no slip" condition

$$\vec{t} \cdot \vec{u} = \vec{t} \cdot \vec{U},$$

where  $\vec{t}$  is tangent to the boundary. Note that our solutions for the Euler equations do not satisfy this condition. Indeed, there are no solutions to it that satisfy it, because they are a first order system and we already imposed "no pass".

## Examples

### Couette Flow



Let  $\frac{\partial \vec{u}}{\partial t} = 0$  and  $\vec{u}(\vec{x}, t) = (u(z), 0, 0)$ . We take the Ansatz

$$(\vec{u} \cdot \nabla) \vec{u} = 0$$

to get

$$\frac{D\vec{u}}{Dt} = 0.$$

There are no pressure gradients, so the NS equation becomes

$$\frac{\partial^2 u}{\partial z^2} = 0$$

with boundary conditions

$$u(0) = 0, \quad u(h) = U.$$

Hence,

$$u(z) = \frac{Uz}{h}$$

Note that

$$\vec{\omega} = \nabla \times \vec{u} = (0, \frac{U}{h}, 0)$$

### Circular Couette Flow, or Taylor-Couette Flow



Here,

$$\vec{u} = \Omega(r)(-y, x, 0)$$

with  $\Omega(R_1) = \Omega_1$ ,  $\Omega(R_2) = \Omega_2$ . We have

$$(\vec{u} \cdot \nabla) \vec{u} = -r\Omega^2 \hat{r},$$

$$\frac{\partial P}{\partial r} = r\Omega^2.$$

Hence,

$$\frac{D\vec{u}}{Dt} = -\nabla P.$$

So any such flow satisfies the Euler equation. For the NS equation, we need

$$\nabla^2 \vec{u} = 0,$$

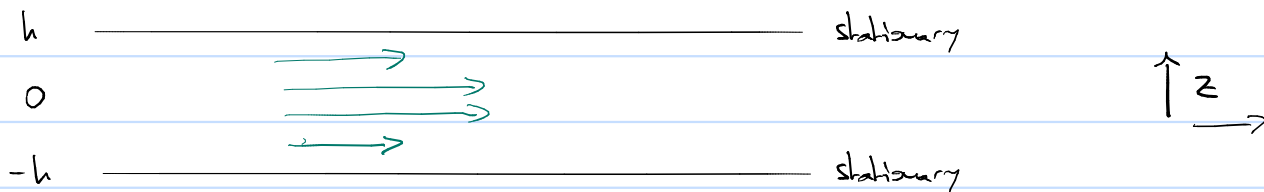
$$\text{i.e.} \quad \Omega'' = -\frac{3\Omega'}{r}$$

$$\text{i.e.} \quad \Omega = A + \frac{B}{r^2}.$$

By the "no slip" conditions,

$$A = \frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}, \quad B = (\Omega_1 - \Omega_2) \frac{R_1^2 R_2^2}{R_2^2 - R_1^2}.$$

### Poiseuille Flow



Here, both plates are stationary, but

$$\frac{\partial P}{\partial x} = \text{const.}$$

Again,  $\vec{u} = (u(z), 0, 0)$ , and the NS equation is

$$\mu \frac{\partial^2 u}{\partial z^2} = \frac{\partial P}{\partial x} = \text{const.}$$

With "no slip", we get

$$u(z) = -\frac{1}{2\mu} \frac{\partial P}{\partial x} (h^2 - z^2)$$

Taking the curl of the NS equation, we get

$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla) \vec{u} + \underbrace{\nu \nabla^2 \vec{\omega}}_{\text{analogous to heat diffusion}} \quad \text{Vorticity equation} \rightarrow \text{viscosity diffuses vorticity}$$

### Burgers Vortex

Let  $\vec{u}(\vec{x}, t) = \vec{u}_{\text{strain}}(\vec{x}) + \vec{u}_{\text{rot}}(\vec{x}, t)$ ,

where

$$\vec{u}_{\text{strain}}(\vec{x}) = \alpha(-x, -y, 2z), \quad \vec{u}_{\text{rot}}(\vec{x}, t) = f(r, t)(-y, x, 0).$$

The rotational part gives rise to the vorticity  $\vec{\omega} = (0, 0, w)$  with

$$w = \frac{1}{r} \frac{\partial}{\partial r}(r^2 f).$$

The vorticity equation gives

$$\frac{\partial w}{\partial t} - \alpha r \frac{\partial w}{\partial r} - 2\alpha w = \nu \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial w}{\partial r} \right).$$

For steady solutions, we get

$$\frac{1}{r} \frac{\partial}{\partial r} \left( \alpha r^2 w + \nu r \frac{\partial w}{\partial r} \right) = 0.$$

Integrating, we get

$$\frac{\partial w}{\partial r} = -\frac{\alpha r}{\nu} w + \text{const.} \quad w \text{ decays at } \infty$$

Hence,

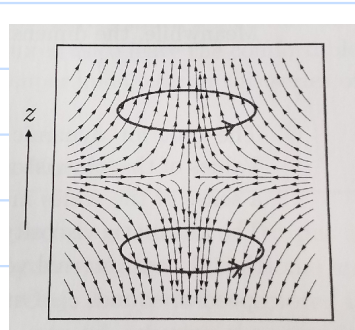
$$w(r) = \frac{\Gamma \alpha}{2\pi \nu} e^{-\frac{\alpha r^2}{2\nu}}.$$

The constant  $\Gamma$  can then also be calculated as the asymptotic circulation of the flow

$$\Gamma = \int_S \vec{\omega} \cdot d\vec{S} = 2\pi \int_0^\infty dr r w(r).$$

Hence, we get the angular velocity profile

$$f(r) = \frac{\Gamma}{2\pi r^2} \left( 1 - e^{-\frac{\alpha r^2}{2\nu}} \right).$$



This **Burgers vortex solution** is the simplest model of a hurricane.

We can calculate the **dissipation** per unit length

$$2\pi \mu \int_0^\infty dr r w^2 = \frac{\Gamma^2 \alpha^2}{4\pi}$$

which is independent of the viscosity  $\nu$